PaperTrail AI

Turn documents into business intelligence

AUTOMATE YOUR WORKFLOW / ROUND 1 PROJECT PROPOSAL

≤ 500 WORDS
10 required responses

01 What does the Skill do?

PaperTrail turns an authorized NAS folder into a source-linked vendor-spend answer, with a clear billed-or-paid basis, evidence, and exceptions.

02 Target user and problem

Small-business owners and bookkeepers keep invoices, receipts, and contracts apart. They need quick answers on bills, payments, and contract compliance.

03 Solution

PaperTrail builds a per-request evidence trail linking each amount to its source and related files. It separates billed from documented-paid amounts, flags potential overlaps or contract differences, and identifies missing bills only against a supplied schedule.

04 Main user flow

User asks, “How much did we spend with Microsoft this year?” and selects an authorized folder. PaperTrail clarifies billed versus paid, verifies files, then returns sourced totals and exceptions; saving requires a request and approval.

05 Difference from existing tools

Unlike a PDF chatbot, PaperTrail connects invoice, payment, and contract evidence across files. Each subtotal has an auditable trail to supporting or conflicting documents; uncertain matches stay visible, not silently deduplicated.

06 UGREEN AI NAS features and steps

`ug-filesystem.find` (`file-agent`) searches the authorized folder. `ug-document.extract`, `qa`, and `reconcile` (`document-agent`) extract, verify, and compare records. Optional `ug-knowledge.save-text` (`knowledge-agent`) saves a requested report after approval; source files stay unchanged.

07 Technical approach

Inputs: question, authorized folder, period, and spending basis. Each request runs fresh scoped discovery; registered experts receive only verified paths and prior results. A separate document-grounded check verifies totals; conflicts or gaps stop a definitive total. The workflow is read-only and needs no scripts, credentials, database, or financial API. Use the NAS-configured model only after verifying provider and data policy. Check the active capability catalog; stop rather than guess if mappings drift.

08 Built so far

The pre-test Skill and synthetic acceptance cases are drafted; the package passes UGREEN’s static checker. NAS execution, permissions, model policy, repeat-use, extraction accuracy, and save approval remain untested.

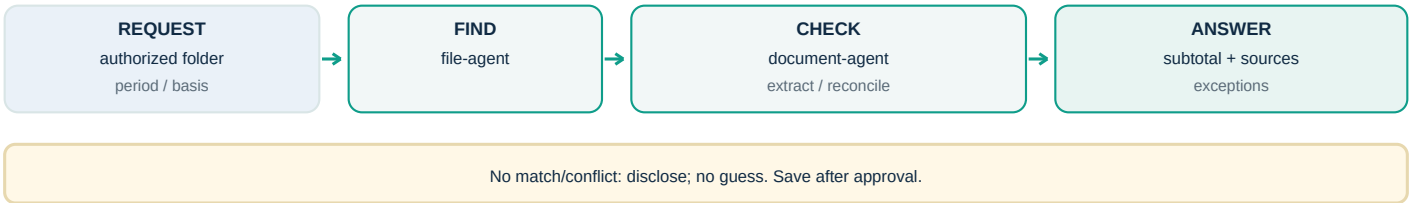
09 Finalist tests

Core 1: total a vendor’s billed invoices with a matching payment receipt; do not double-count it or infer another invoice was paid. Core 2: reconcile invoice and contract, compare a supplied billing schedule, and exclude unrelated files; flag differences and report expected bills only as “not found here.” Edge: an unconfirmed invoice/receipt match or unreadable amount must yield uncertainty, not a guessed total.

10 Limits, risks, dependencies

Poor scans, vendor ambiguity, tax, or mixed currencies impair extraction. Files cannot prove absent facts or fraud. No payment approval, accounting updates, or monitoring. Runtime depends on UGREEN agents, permissions, and approved model; NAS testing is pending.

WORKFLOW



DISCLOSURES

No third-party code is bundled. UGREEN Skill Creation Tool v0.7.3 and catalog were consulted, not redistributed; no separate license notice was found in the archive.

Sources: Devpost · UGREEN docs

Pre-test build · NAS validation pending